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Palm OS® Support for Expansion and Secondary Storage—Part I

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Conference 2000

Last Week…

- 8MB was a lot of data!!!
  - About 100 applications (!!!)
  - “The Count of Monte Cristo” in Doc format
  - 1400 addresses, 900 appointments, etc.

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  - About 100 applications (!!!)
  - “The Count of Monte Cristo” in Doc format
  - 1400 addresses, 900 appointments, etc.
- …But some applications have suffered!

This Week…

High-capacity Removable Media









- Pictures…
  - … of your extended family!
  - … of your vacation!
  - … of your house!
  - … of your pets!
  - … from PalmSource™!

This Week…

High-capacity Removable Media









- Maps of…
  - … your entire home state in detail!
  - … your country!
  - … the whole world!

This Week…

High-capacity Removable Media









- Documents…
  - Novels!
  - Reference materials!
  - Work in progress!
  - The entire Project Gutenberg library!!!

This Week…

High-capacity Removable Media



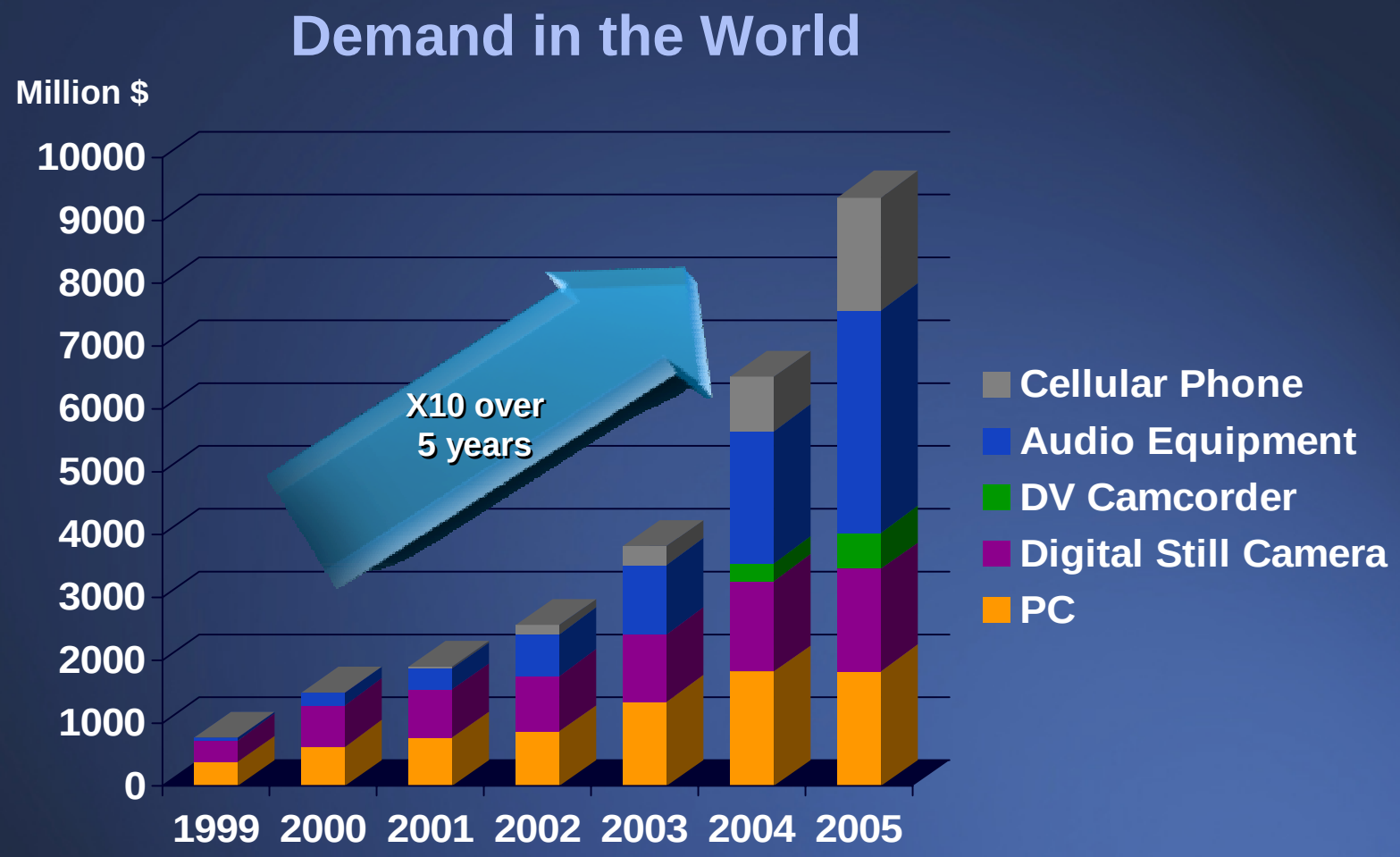






- Games with...
  - … 16-bit graphics!
  - … sampled sounds!
  - … extensive levels!

World Demand for Memory Cards



Source: Technology System Research

Interoperability of Memory Cards



Audio & Visual

TV/DVC/DSC
Set Top Box
Portable Audio Equipment
DVD Audio Equipment

Mobile

Cellular/PHS Phone/PDA
Next Generation Mobile
Communication Terminal
Car Navigation System
Car Audio Equipment





Compact

PC

Personal Computer Printer
PC Card Adapter
Scanner

Large Capacity



Secure

Telecommunications

Internet Phone
Cordless Phone
Fax/E-Mail Terminal

High Speed



Entertainment Game, etc.

KIOSK Terminal
Health Care and
Medical Equipment

Home Appliance

Microwave Oven
Rice Cooker/Refrigerator
Air Conditioner
Air Purifier







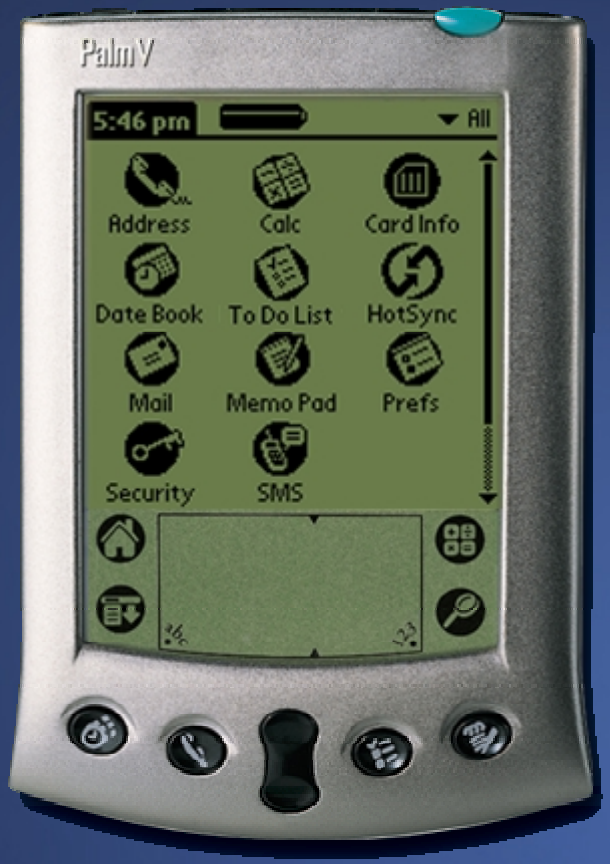
Palm OS® Expansion Highlights

- Hot insertion and removal of multiple cards
- Architecture can support various media types
- Support for the FAT 12/16 filesystem standard
- Basic support for custom I/O applications

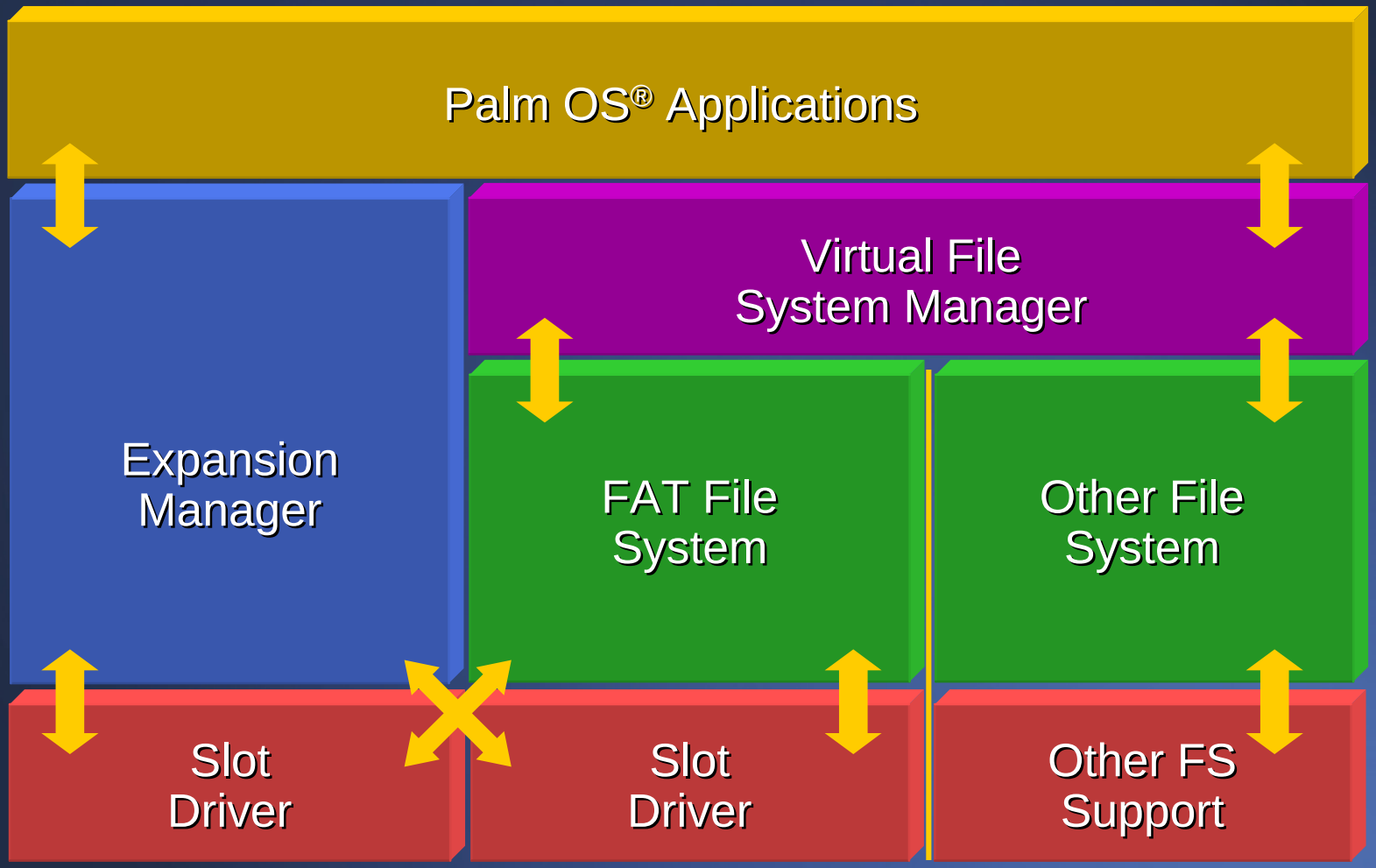
Palm OS® Expansion Highlights

- Extends HotSync® APIs provide access to expansion filesystem
- Support for an “auto start” application
- Modular system extension, not always present
- Integrated with Palm OS® Launcher

Launcher Demo



Functional Block Diagram



Expansion Manager



- Provides services to slot drivers:
  - Install slot drivers during boot
  - Slot number allocation
  - Interrupt safe card insertion and removal notification
  - Automatically mount volumes on card insertion
- Provides services to applications:
  - Slot enumeration and status
  - Card inserted and removed notifications
  - Detailed card information
  - Direct access to slot drivers

Virtual Filesystem Manager

Virtual File
System Manager

- Manages all installed filesystems
- Load and execute “auto start” apps, if present
- Broadcast volume mount notifications
- Provides a unified API for file access

VFS API Overview

- Manipulate Files:
  - Open & Close files: ***VFSFileOpen(...), …***
  - Read & Write files: ***VFSFileRead(...), …***
  - Create, Delete, & Rename files
  - Directory Enumeration
- Manipulate Volumes
  - Format, Label, Mount, Unmount

VFS API Overview

- Manipulate Files:
  - Open & Close files: ***VFSFileOpen(...)***
  - Read & Write files: ***VFSFileRead(...)***
  - Create, Delete, & Rename files
  - Directory Enumeration
- Manipulate Volumes
  - Format, Label, Mount, Unmount
- Volume Enumeration
  - ***VFSVolumeEnumerate(...)***

VFS Utility APIs

- Import/Export PRC or PDB files:
  - ***VFSImportDatabaseFromFile(…)***
  - ***VFSExportDatabaseToFile(…)***
- Access info or records from PRC or PDB files:
  - ***VFSFileDBInfo(…)***
  - ***VFSFileDBGetRecord(...)***
  - ***VFSFileDBGetResource(...)***

VFS Directory Mapping APIs

- Default locations for certain file types
  - ***VFSGetDefaultDirectory(…)***
  - ***VFSRegisterDefaultDirectory(…)***
  - ***VFSUnregisterDefaultDirectory(…)***

- …Each media type has its own standard! :-(
  - /DCIM vs. /IMAGES for pictures

FileSystem Directory Layout

Directory	Description
/	Root of volume
/PALM	All data written by any Palm OS® application or OS component
/PALM/start.prc	Auto initialization application, optional
/PALM/Backup	Used by Palm OS for backup purposes
/PALM/Launcher	Launcher visible applications
/PALM/Programs	Catch all for other applications and data Your apps should create directories here

Before Using VFS APIs

Check for Presence of VFS Manager!

Sample:

```
err = FtrGet(sysFileCVFSMgr, vfsFtrIDVersion,
                  &vfsMgrVersion);

if(!err) {
      // VFS Mgr is Present
      // Check version of VFS Manager, if necessary
}
else  {
      // VFS Mgr is not installed.
}
```

Code Sample #1

Finding a volume

```
UInt32 iterator = vfsIteratorStart;
UInt16 volRefNum;

while(VFSVolumeEnumerate(&volRefNum, &iterator)
              == errNone)
{
      … do something to this volume …
}
```

Code Sample #2

Creating a text file

```
FileRef     ref;
UInt16      size;

VFSFileCreate(volRefNum, “/PALM/jabberwocky.txt”);
...
```

Code Sample #2

Creating a text file

```
FileRef     ref;
UInt16      size;

VFSFileCreate(volRefNum, “/PALM/jabberwocky.txt”);
VFSFileOpen(volRefNum, “/PALM/jabberwocky.txt”,
            vfsModeReadWrite, &ref);
...
```

Code Sample #2

Creating a text file

```
FileRef     ref;
UInt16      size;

VFSFileCreate(volRefNum, “/PALM/jabberwocky.txt”);
VFSFileOpen(volRefNum, “/PALM/jabberwocky.txt”,
            vfsModeReadWrite, &ref);
size = StrLen(jabberwockyStringP);
VFSFileResize(ref, size);
...
```

Code Sample #2

Creating a text file

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FileRef     ref;
UInt16      size;

VFSFileCreate(volRefNum, “/PALM/jabberwocky.txt”);
VFSFileOpen(volRefNum, “/PALM/jabberwocky.txt”,
            vfsModeReadWrite, &ref);
size = StrLen(jabberwockyStringP);
VFSFileResize(ref, size);
VFSFileWrite(ref, size, jabberwockyStringP, NULL);
...
```

Code Sample #2

Creating a text file

```
FileRef     ref;
UInt16      size;

VFSFileCreate(volRefNum, “/PALM/jabberwocky.txt”);
VFSFileOpen(volRefNum, “/PALM/jabberwocky.txt”,
            vfsModeReadWrite, &ref);
size = StrLen(jabberwockyStringP);
VFSFileResize(ref, size);
VFSFileWrite(ref, size, jabberwockyStringP, NULL);
VFSFileClose(ref);
```

Code Sample #3

Directory Enumeration

```
UInt32              dirEntry;
FileInfoType        fInfo;
FileRef             dirRef;

fInfo.nameP = MemPtrNew(30);
fInfo.nameBufLen = 30;
VFSFileOpen(volRefNum, “/Palm/Launcher/”,
                    vfsModeRead, &dirRef);
...
```

Code Sample #3

Directory Enumeration

```
UInt32              dirEntry;
FileInfoType        fInfo;
FileRef             dirRef;

fInfo.nameP = MemPtrNew(30);
fInfo.nameBufLen = 30;
VFSFileOpen(volRefNum, “/Palm/Launcher/”,
                    vfsModeRead, &dirRef);

dirEntry = vfsIteratorStart;
while(VFSDirEntryEnumerate(volRefNum, dirRef,
                    &dirEntry, &fInfo) == errNone)
{
      AddFile(fInfo.nameP);
}
```

Summary

- Powerful architecture, incredible possibilities!

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- Powerful architecture, incredible possibilities!
- Strive for simple, transparent technology!

Summary

- Powerful architecture, incredible possibilities!
- Strive for simple, transparent technology!
- We've laid the groundwork…
  - But we need you to help us build the rest!


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Q&A


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